

[!IMPORTANT]

STOP ALL WARS

Non gridate più

Cessate d'uccidere i morti,
Non gridate più, non gridate
Se li volete ancora udire,
Se sperate di non perire.

Hanno l'impercettibile sussurro,
Non fanno più rumore
Del crescere dell'erba,
Lieta dove non passa l'uomo.

— *Giuseppe Ungaretti*

Millennial civilizations and sovereign nations are being razed to the ground by the same logic of dominance that has stained human history for centuries. From the Middle East to Ukraine, we are witnessing the destruction of our common heritage. No modern ideology or national interest can justify these crimes.

Silence is complicity, stop the massacre.

oa-tools - A Next-Generation Remastering Suite

After years of passionate work on **penguins-eggs**, I am now undertaking its next evolution: **oa-tools**. oa-tools is built on **oa**, a high-performance C-native remastering engine, and **coa**, its intelligent Go orchestrator. This new architecture replaces fragile bash scripts with native Linux kernel syscalls and strict zero-copy principles — delivering a significant leap in speed and reliability. The goal is ambitious: a truly universal approach to Linux remastering. Every distribution is different, yet they all share common roots. oa-tools already supports Alpine, Arch, Debian, Fedora, Manjaro and openSUSE — and aims to prove that a single, unified engine can handle them all.



oa-tools Architecture

We have transitioned to a monorepo structure to ensure perfect synchronization between the workhorse (oa) and the orchestrator (coa).

coa (**brooding in my dialect**) - The Mind

It manages the full lifecycle: from laying the ISO to the final installation.

The name derives from the dialect word *coa*, referring to the act of brooding or incubating eggs until they are ready to hatch.

Language: Go

For *coa* commands, see [coa command Reference](#).

oa (eggs in my dialect) - The Workhorse

Language: C *oa* is the low-level engine. It handles the "heavy lifting" of the system:

- Managing OverlayFS and mount points.
- Executing SquashFS compression.
- Remove and create users and group via `yocto_style` functions.
- Interacting directly with the Linux Kernel and system binaries.
- **Philosophy:** Performance, stability, and zero-dependency execution.

Getting Started

Prerequisites

- A Linux system (Debian-based, Arch-based. Fedora-based or Manjaro-based).
- `gcc` and `make` (for *oa*).
- `golang` 1.21+ (for *coa*).

Build Everything

From this root directory, simply run:

```
make
```

This will compile both binaries:

- `./oa/oa` (The Engine)
- `./coa/coa` (The Orchestrator)

TIP: Actually you can get native packages, simply using `make clean package`

Philosophy

The **oa-tools** project aims to provide a "Passepartout" for Linux remastering. By separating the **Mind** (Go) from the **Workhorse** (C), we achieve a clean, maintainable, and incredibly fast workflow that can adapt to any distribution without changing the user experience.

Documentation about *oa-tools* can be found on [DOCS](#).

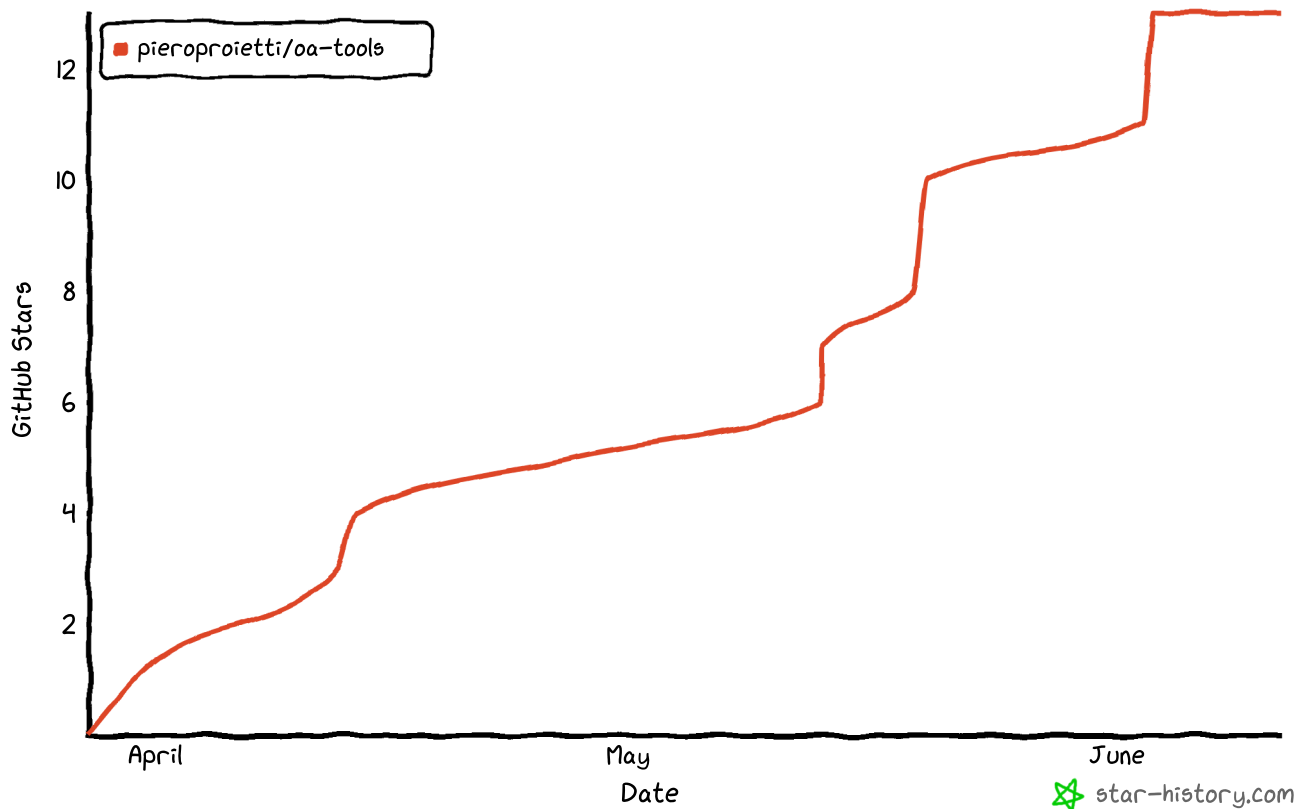
The article [eggs-bananas](#) philosophy can be read on my blog.



Star History

This project collects stars, look to the sky... contribute!

Star History



Created with passion by Piero Proietti.

blog

- <https://penguins-eggs.net>